Galen 2019 AML — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

19/28 samples (≥31 cells) intermix on one continuous manifold with no per-patient islands (iLISI 5.66 of 9.95 ceiling; radius-graph: 1 detached); both lobes (Cycling 9,872 / Low-cycling 6,227) carry the trajectory.

Gate 2 — module geography → GO (non-branching, inversion FLAG)

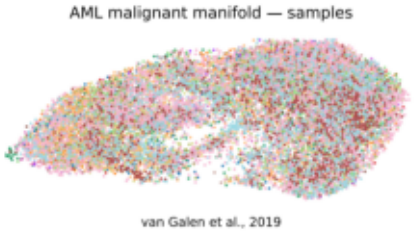
Gast/pSom cross-repress (r=−0.67); along pseudotime pSom declines / Gast rises (ρ=−0.87/+0.44); Crest anti-correlated with pSom (r=−0.90) but positive with Gast (r=+0.39) — the inversion — rising to a terminal readout (ρ=+0.83; crest-high decile mean pt=0.99).

Gate 3 — cross-repression → NA

Single monotone axis — no bifurcation and no Gast/pSom co-expression hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

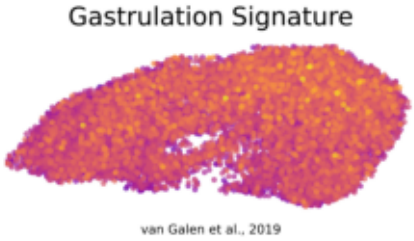
Manifold — samples



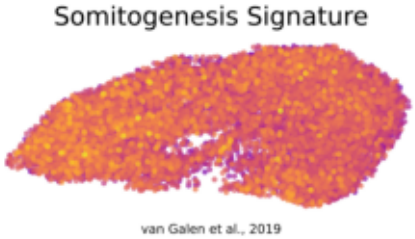
Pseudotime



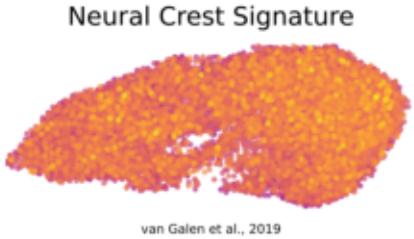
Gastrulation



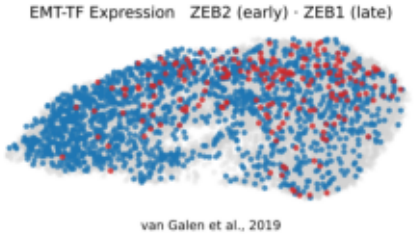
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Acute Myeloid Leukemia (van Galen et al., 2019), SeqWell — 37 samples / 33,392 cells (master scope); malignant only: 16,169 load, 16,099 across 19 samples (≥31) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Galen's AML malignant manifold is a single monotone Gast/pSom (cross-repressing) → terminal-Crest gradient rooted at the primitive, pSom/Gast-high, low-cycling pole; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The developmental axis spans both lobes (pSom pole low-cycling, Crest terminus cycling), so the whole manifold is kept (Bischoff/Sun/Lee no-scoping precedent). FLAG: Crest enriches the Gast end (inversion). EMT-TFs ZEB2 (early/root) → ZEB1 (late/terminus); FOXC1/SNAI1/TWIST below 3% floor.

Parameter choices

Min cells/sample	31 (keeps 19/28; cliff 61→16)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	tip 2 = Gast/pSom-high, Crest-low	**Compartments**	Cycling 9,872 / Low-cycling 6,227
Geometry	Linear (non-branching); inversion FLAG	**EMT-TFs**	ZEB2 (early/root) · ZEB1 (late/terminus)